GLOBAL ENVIRONMENTAL STANDARDS & DATASETS

An implementation guide for evidence-based air-quality, water-quality and water-body pollution explanations

For: Environmental Intelligence & Compliance Management System

Coverage: International guidance, regional laws, national reference systems, monitoring datasets, geospatial data and explainable AI design

Research cut-off: 03 November 2026

Prepared as a practical research and software implementation reference

Read this before deploying the global feature

This report is a curated global implementation guide for a software product. It provides a global health-guidance layer, examples of major regional and national regulatory systems, official monitoring data portals, legal discovery repositories, geospatial layers and a recommended evidence-engine design. It is not an exhaustive legal register of every country, state, province, pollutant, industrial sector, local permit or water-body designation. A production compliance verdict must be supported by the current primary law and, where relevant, the site-specific permit that applies to the place and sample date.

Core rule: WHO guidance is not automatically law in every country. UNEP SDG 6.3.2 is a global monitoring framework whose “good” targets are set nationally. Regional / national laws and site-specific permits determine legal compliance. If no applicable law has been resolved, label the result “guideline comparison only”.

Four distinctions your website must never collapse

Do not confuse	Correct treatment
AQI and pollutant concentration	AQI is a jurisdiction-specific index. Store and display the underlying pollutant concentrations separately.
Ambient water and effluent	Ambient river/lake criteria assess receiving water; effluent limits assess a discharge at a defined outfall. They may use different methods and averaging periods.
Legal limits and guidance	A regulation or permit may be enforceable. WHO / IFC / CCME / ANZG guidance may be advisory or adopted differently in local law.
District and watershed	Districts are administrative areas; water moves through catchments and river reaches. Keep both spatial identities.
Measured and modelled data	Ground samples, sensor feeds, satellite retrievals and model forecasts are different evidence types. Label their provenance and limitations.

Current-version changes checked for this edition

- WHO drinking-water: a revised WHO Guidelines for Drinking-water Quality edition incorporating the third addendum was published on 17 June 2026.
- China ambient air: GB 3095-2026 took effect on 1 March 2026 and replaced GB 3095-2012.
- European water quality: Directive (EU) 2026/805, dated 30 March 2026, amended the Water Framework, Groundwater and Environmental Quality Standards directives.
- Global freshwater data: GEMStat publishes an open-data subset through Zenodo, while the portal announced temporary service unavailability dated 7 October 2026. Check live status before production integration.

Research cut-off: 03 November 2026. Legal texts, data availability, APIs and dataset terms can change. Refresh and review the official sources before relying on this guide for compliance decisions.

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Recommended implementation principle: store measurements in their original units and metadata; select the law or guideline only after resolving the jurisdiction, environmental medium, water use, averaging period and measurement method. AI should explain a deterministic standards comparison, not invent one.

How to read status labels in this report

Label	Meaning in your software
Binding legal standard	A requirement in legislation, regulation, permit, consent condition or formally adopted instrument applicable to the location and activity.
Guideline / health-based recommendation	Scientific or policy recommendation that may inform public-health interpretation but is not automatically a legal violation threshold.
Monitoring classification	A categorisation produced by a monitoring programme (for example river-priority bands or AQI categories). Do not mislabel it as a statutory permit limit.
Modelled / satellite estimate	A spatial estimate, forecast or proxy. Useful for context and screening; not automatically equivalent to an accredited ground sample.
Citizen-reported / unverified	User supplied data without independent validation. It can trigger review but must remain clearly labelled.

1. Scope, terminology and legal hierarchy

The platform should classify the measurement before it searches for a threshold.

1.1 Five different assessment problems

Assessment type	What is measured	What it must be compared with
Ambient air	Outdoor concentration at a monitor or validated sensor (PM2.5, PM10, NO2, O3, SO2, CO, etc.)	Local ambient-air standard and/or WHO health-based guideline. Calculate AQI only using the local official formula.
Source / stack emissions	Concentration, mass rate, opacity or process-specific emission from a chimney, engine or facility	Permit, sector-specific emission standard, monitoring method and operating conditions.
Ambient surface water	River, lake, reservoir, estuary, coastal or marine sample	Applicable designated use, water-body classification, chemical/ecological objective or local water-quality criteria.
Drinking water	Water intended for human consumption at source, treatment plant or consumer tap	National drinking-water regulation; use WHO guidance as a reference where appropriate.
Wastewater / effluent	Discharge from municipal STP, industrial unit, mine, farm or other point source	Permit/consent and applicable sectoral or general effluent standard for the receiving medium.

1.2 Legal and scientific hierarchy for rule selection

- 1. Site-specific permit or consent: facility-specific permit limits, monitoring conditions, discharge point, operating conditions and enforceable orders. These are often the most directly applicable requirements for a named facility.
- 2. Local and national law: current national / state / provincial legislation, regulations, ambient standards, water-body objectives, sector rules and official technical methods.
- 3. Regional law: for example EU directives and regulations, but only as implemented/applicable in the member jurisdiction and on the relevant date.
- 4. International health / technical guidance: WHO, UNEP, FAO and technical bodies provide reference values and methods. They are not automatically domestic law.
- 5. Research / model / satellite context: use to investigate patterns and prioritise field verification; do not use as a substitute for applicable legal limits.

Never select a threshold from country name alone. Resolve country, state/province, district, river basin, water-body segment, environmental medium, intended use, source type, sampling date, standard version and legal status.

1.3 Why “district-wise” is a data-model problem

Districts are administrative reporting units; rivers and lakes follow hydrological boundaries. One river may cross several districts, and one district may contain several river basins. Keep both administrative geography and hydrological geography in the database. Store the station coordinate and official water-body / reach identifier, then derive district for display. Do not assign a single pollution label to an entire district from one sample.

2. International reference frameworks

These establish common methods and reference concepts, but their legal force varies.

2.1 World Health Organization (WHO)

• WHO Global Air Quality Guidelines (2021): health-based levels for PM2.5, PM10, ozone, NO2, SO2 and CO. Use for health interpretation and trend context. They are recommendations, not a universal legal code. Official guideline.

• WHO Guidelines for Drinking-water Quality: fourth edition incorporating first, second and third addenda, with a further update published 17 June 2026. Use the current 2026 edition and its chemical, microbial, radiological and operational guidance rather than copying an old spreadsheet. It promotes water safety planning from catchment to consumer and independent surveillance. Latest WHO drinking-water guideline (2026).

2.2 United Nations Environment Programme (UNEP) - SDG 6.3.2

UNEP SDG Indicator 6.3.2 measures the proportion of water bodies with good ambient water quality. The global method uses five core parameter groups: dissolved oxygen (surface water), electrical conductivity (surface and groundwater), nitrogen/nitrate (surface and groundwater), phosphorus (surface water) and pH (surface and groundwater). The key design point is that “good” is set at the national level; this is a comparable monitoring framework, not a single worldwide concentration limit. UNEP SDG 6.3.2.

2.3 UNEP GEMS/Water and GEMStat

GEMS/Water supports global freshwater monitoring and data exchange. GEMStat includes monitoring data shared by countries and organisations, with station, country and catchment-level access. Its catalogue describes more than 600 parameters. Data availability and license vary by provider; keep provenance and provider attribution. The portal documentation says open-licensed subsets are available through an annual Zenodo dataset; larger data requests may be needed for global extracts. GEMStat portal | GEMStat data availability.

Service notice observed during research: the GEMStat portal announced temporary unavailability from a DDoS incident dated 07 October 2026. Check the live status before building a production dependency; use the Zenodo open-data mirror where its license and coverage fit your use case.

2.4 Other international technical frameworks

Framework	Use in platform	Legal status caution
FAO water / irrigation resources	Irrigation water, agriculture, water stress and water-use context. FAO AQUASTAT.	AQUASTAT is country water-resource / agriculture information, not a substitute for local contaminant sample results.
IFC / World Bank Group EHS Guidelines	Industrial air emissions, wastewater and ambient water quality, water conservation, waste and monitoring; pair general EHS with the relevant sector guideline. General EHS Guidelines.	Good International Industry Practice benchmark; not automatically a local permit or law.
UN SDG 11.6.2 / WHO air database	City-level annual PM2.5, PM10 and NO2 data for global comparison. WHO Air Quality Database.	Not a live station feed; city-level annual means, coverage and update cycle must be shown.
International environmental treaties	Transboundary water, marine pollution, hazardous waste and persistent pollutants context.	Treaty obligations need domestic implementation and may not provide site-specific concentration thresholds.

3. Air quality: global health guidance and regional law

Store pollutant concentration and AQI separately. A single AQI formula is not global.

3.1 WHO 2021 Global Air Quality Guideline levels

The WHO values below are health-based guideline levels. The short-term values are 99th-percentile concentrations (approximately 3-4 exceedance days per year); ozone peak-season definition is based on six consecutive months with the highest six-month running average of daily maximum 8-hour ozone.

Pollutant	Averaging time	WHO AQG level	Interim targets (IT-1 / IT-2 / IT-3 / IT-4)
PM2.5	Annual	5 µg/m³	35 / 25 / 15 / 10 µg/m³
PM2.5	24-hour	15 µg/m³	75 / 50 / 37.5 / 25 µg/m³
PM10	Annual	15 µg/m³	70 / 50 / 30 / 20 µg/m³
PM10	24-hour	45 µg/m³	150 / 100 / 75 / 50 µg/m³
Ozone (O3)	Peak season	60 µg/m³	100 / 70 / - / - µg/m³
Ozone (O3)	8-hour	100 µg/m³	160 / 120 / - / - µg/m³
Nitrogen dioxide (NO2)	Annual	10 µg/m³	40 / 30 / 20 / - µg/m³
Nitrogen dioxide (NO2)	24-hour	25 µg/m³	120 / 50 / - / - µg/m³
Sulfur dioxide (SO2)	24-hour	40 µg/m³	125 / 50 / - / - µg/m³
Carbon monoxide (CO)	24-hour	4 mg/m³	7 / - / - / - mg/m³

Source: WHO 2021 guideline summary table. Use the exact averaging and percentile method; never compare a momentary sensor value directly with a 24-hour or annual guideline. WHO guideline | WHO table in NCBI Bookshelf

3.2 AQI is jurisdiction-specific

AQI is a public communication index calculated from pollutant-specific sub-indices, averaging periods, breakpoint tables, minimum data completeness and the jurisdiction's own rules. The U.S. AQI, Indian National AQI, EU Air Quality Index and other national systems are not interchangeable. If the platform serves multiple countries, label the index with its source methodology (e.g., "US EPA AQI", "India CPCB AQI"), not simply "AQI". If no local AQI method applies, show concentrations and WHO guideline comparison instead.

3.3 Examples of national / regional air law

Jurisdiction	Instrument / status	Implementation point
United States	Clean Air Act; US EPA NAAQS. The 2024 annual primary PM2.5 standard was revised to 9.0 µg/m³; 24-hour primary PM2.5 standard is 35 µg/m³ (98th percentile, 3-year form).	Use EPA current NAAQS table and local monitor/designation data. NAAQS form and multi-year statistics matter; one reading does not establish nonattainment. EPA NAAQS table.
European Union	Directive (EU) 2024/2881. New health-based limit values are to be attained by 1 Jan 2030. Examples: annual PM2.5 10 µg/m³, daily PM2.5 25 µg/m³ (max 18 days); annual PM10 20 µg/m³, daily PM10 45 µg/m³ (max 18 days); annual NO2 20 µg/m³; annual SO2 20 µg/m³.	The directive also contains an earlier transitional table with limit values to be attained by 11 Dec 2026. On the research cut-off date (3 Nov 2026), do not use the 2030 values as though they are already the current 2026 limit values. Resolve transposition, applicable date and any derogation. Directive 2024/2881.
China	Ambient Air Quality Standard GB 3095-2026. Effective 1 Mar 2026; replaced GB 3095-2012 and is mandatory under the issuing notice.	Retrieve the official Chinese standard text / annex tables for pollutant-specific limits, functional zones, averaging times, methods and data-validity rules. Do not reuse older GB 3095-2012 values. China MEE announcement; GB 3095-2026 standard page.
Japan	Environmental Quality Standards of Japan for air, water, soil and noise.	Use the Ministry of Environment's current standards and revision history. Japan standards portal.
Canada	Canadian Ambient Air Quality Standards and National Air Pollution Surveillance data systems.	Load the current Canadian Council of Ministers / federal standard values from official federal sources; NAPS is the measurement dataset, not the threshold source. Canada NAPS.
Australia	National Environment Protection (Ambient Air Quality) Measure and state/territory implementation.	Store the applicable measure version, pollutant, averaging period and exceedance form from the current Federal Register / NEPM source.

3.4 Air assessment algorithm must respect statistical form

- Identify pollutant, instrument, monitor type, station, timestamp, unit and averaging window.
- Validate data capture, calibration / method status and completeness requirements specified by local law.
- Calculate the statistic required by law: hourly maximum, daily mean, annual mean, percentile, number of permitted exceedance days or multi-year design value.
- Compare with the applicable local legal standard and separately compare with WHO guideline if useful.
- Calculate local AQI only from complete, valid observations using the relevant authority’s official breakpoints and formula.
- Return separate fields for legal-standard assessment, health-guideline comparison and AQI category.

4. Water quality: ambient freshwater, marine water and drinking water

Do not combine ambient receiving-water criteria, drinking-water limits and effluent discharge limits.

4.1 Global ambient freshwater assessment: SDG 6.3.2

Core parameter	Medium	What it helps explain	Caution
Dissolved oxygen (DO)	Surface water	Oxygen available for aquatic organisms; low DO may be consistent with organic loading / eutrophication.	Interpret with temperature, flow, time of day, sampling method and ecosystem context.
Electrical conductivity (EC)	Surface and groundwater	Dissolved ionic content / salinity proxy; useful for change detection.	Not a direct measure of a specific toxic pollutant.
Nitrogen / nitrate	Surface and groundwater	Nutrient loading and possible contamination pathways.	Record chemical form and reporting basis (as N, NO3 or NO3-N).
Phosphorus	Surface water	Nutrient enrichment and eutrophication pressure.	Distinguish total phosphorus, orthophosphate and dissolved fractions.
pH	Surface and groundwater	Acid-base condition and potential stress on aquatic life / treatment processes.	Check calibration, temperature and method.

UNEP's SDG 6.3.2 framework uses these five core parameters, but "good ambient water quality" is assessed against national targets and water-body definitions. It does not establish one global universal BOD, COD, nitrate or phosphorus limit. UNEP SDG 6.3.2 methodology and data.

4.2 Water-body types must be explicit

Water-body / use	Required context before comparing values
River / stream	Water-body ID, upstream/downstream station, flow regime, designated use, season, sediment conditions and catchment pressures.
Lake / reservoir	Lake type, depth / stratification, trophic state, intake/recreation use, seasonal bloom conditions and representative sampling depth.
Groundwater	Aquifer, well depth, intended use, redox state, geology and whether the result is raw groundwater or treated supply.
Estuary / coastal / marine	Salinity zone, tidal phase, water-body class, bathing / shellfish / mariculture use, sediment and seasonal conditions.
Drinking water	Raw, treated, distribution or point-of-consumption sample; supply system and national potable-water standard.
Wastewater discharge	Source type, discharge point, receiving medium, permit/consent, treatment stage, flow and sampling method.

4.3 WHO drinking-water guidance - use the latest edition

WHO published the fourth edition incorporating the first, second and third addenda on 17 June 2026. This is the current reference edition located during research. It is a risk-management framework covering health-based targets, water safety plans, catchment-to-consumer controls and independent surveillance. For a production database, ingest the current PDF and accompanying tables, record each parameter's guideline value, unit, provisional status, derivation notes and effective publication date, and keep earlier editions for historical interpretation. WHO 2026 Guidelines for Drinking-water Quality.

Do not present WHO guidance as a legally enforceable potable-water limit in every country. The country's drinking-water regulations and utility rules control the compliance result. WHO values are a global health reference unless adopted by local law.

4.4 Examples of drinking-water legal values in a regional system

EU Drinking Water Directive 2020/2184 sets parametric values for water intended for human consumption. Its microbiological values include E. coli = 0 number/100 mL and intestinal enterococci = 0 number/100 mL (bottled/container water has a separate 250 mL sample volume). The directive also sets chemical parameters, including arsenic = 10 µg/L. These are EU legal values, not a global standard. Check the current consolidated text and national transposition. Directive (EU) 2020/2184.

4.5 Surface-water legal frameworks: regional examples

Jurisdiction	Framework	How to use it
European Union	Water Framework Directive 2000/60/EC; Environmental Quality Standards Directive 2008/105/EC; Groundwater Directive 2006/118/EC; updated by Directive (EU) 2026/805.	Assessment aims to prevent deterioration and achieve good ecological and chemical status. Use water-body classifications, river basin management plans, monitoring and substance-specific EQS. 2026/805 amends WFD, groundwater and EQS rules; store new version and transition dates. EU WFD; Directive 2026/805.
United States	Clean Water Act; EPA recommended national water-quality criteria; state/tribal standards and discharge permits.	EPA Section 304(a) criteria are recommended guidance for states/tribes to develop standards; they are not in every case a direct permit limit. Use local water-quality standards and NPDES permit limits for enforceable assessments. EPA recommended criteria.
Canada	Canadian Environmental Quality Guidelines (CCME), plus federal/provincial/territorial standards.	Choose freshwater/marine aquatic life, irrigation, livestock, sediment, soil or groundwater guideline as appropriate. Guidelines are science-based and implementation varies by jurisdiction. CCME guidelines.
Australia / New Zealand	Australian and New Zealand Guidelines for Fresh and Marine Water Quality (2018 online revision).	Use ecosystem type, region, ecosystem condition, level of protection and default guideline values. The framework uses weight-of-evidence and toxicant-specific technical briefs; not every default value is automatically law. ANZG water guidelines.
Japan	Environmental Quality Standards for water and groundwater.	Use current MOE parameter tables and water-body type / designated use. Japan standards.
China	Surface Water Environmental Quality Standard GB 3838-2002 (shown as current on SAMR standard page during research).	It applies to usable surface waters including rivers, lakes, canals, channels and reservoirs and has use/function-based classes. Retrieve official Chinese text and amendments. MEE English description; SAMR standard status.
India	CPCB designated-best-use criteria, Water Act and Environment (Protection) Rules; state PCB consent conditions.	Use surface-water designated use, discharge destination, sectoral standards, local consent and current amendments. See your India implementation reference for the CPCB A-E criteria and Maharashtra MPCB sources.

5. Wastewater, effluent and industrial discharge

There is no single global effluent table that can replace a facility permit and sector rule.

5.1 Correct comparison sequence

- Identify whether sample is untreated sewage, treated sewage, industrial wastewater, cooling water, stormwater, mine drainage, landfill leachate, aquaculture discharge or another stream.
- Identify discharge destination: inland surface water, sewer, land, coastal/marine water, groundwater injection or reuse.
- Retrieve discharge permit / consent and all applicable national and subnational standards.
- Check sector-specific rules before falling back to general standards.
- Match analytical method, sample type (grab/composite), test duration, temperature and reporting basis.
- Compare concentration and, where required, pollutant load (concentration × flow); verify the permit's averaging period, sample frequency and maximum/average form.

5.2 International benchmark: IFC / World Bank Group EHS Guidelines

The General EHS Guidelines cover air emissions and ambient air quality, wastewater and ambient water quality, water conservation, waste management and monitoring. They are meant to be read together with the relevant industry-sector EHS guideline. Use this as an international good-practice benchmark for screening and project design, not as a direct legal limit unless a lender agreement, permit or contract makes it applicable. The World Bank Group states that its EHS Guidelines are being updated. IFC General EHS Guidelines.

5.3 Municipal wastewater: European Union example

The recast Urban Wastewater Treatment Directive (EU) 2024/3019 sets a modernised framework for collection and treatment, including staged requirements for tertiary nutrient treatment and quaternary treatment for micropollutants. For larger plants (150,000 population equivalent and above), the directive stages quaternary treatment coverage: 20% by 31 Dec 2033, 60% by 31 Dec 2039 and all such discharges by 31 Dec 2045. These are scheduled legal obligations for EU member states, not immediate concentration thresholds applicable worldwide. Directive (EU) 2024/3019.

5.4 Industrial emissions and pollutants released

Use regulatory facility registries and pollutant release and transfer registers (PRTRs) to identify potential sources, permit status and historic releases. These are source-pressure datasets, not proof that a facility caused a particular downstream measurement. Spatial proximity alone is not causal evidence. The EU Industrial Emissions Portal provides facility locations and reported releases / transfers to air, water and land for major industrial sites. EU Industrial Emissions datasets.

6. Regional and national regulatory reference map

Use this as a discovery map. Confirm the controlling instrument in the official national gazette or consolidated legal database.

Region / country	Air standards source	Water / discharge source	Official legal discovery
Global	WHO AQG; WHO ambient air database	WHO drinking water; UNEP SDG 6.3.2; GEMS/Water	WHO / UNEP / FAOLEX / ECOLEX
EU / EEA	Directive 2024/2881; EEA air quality e-reporting	WFD, EQS, groundwater and drinking-water directives; Waterbase / WISE	EUR-Lex; European Commission; EEA
USA	Clean Air Act / EPA NAAQS; AQS / AirNow	Clean Water Act; EPA criteria; state standards; NPDES permits	US Code / eCFR / EPA
Canada	CAAQS; federal/provincial standards; NAPS data	CCME CEQGs; federal/provincial standards	Justice Laws; ECCC; CCME
Australia / New Zealand	Ambient Air NEPM and local state requirements	ANZG 2018; national/state water statutes and licences	Federal Register of Legislation; waterquality.gov.au
China	GB 3095-2026 (effective 1 Mar 2026)	GB 3838-2002 surface-water standard and other water rules	Ministry of Ecology and Environment; SAMR standards portal
Japan	MOE Environmental Quality Standards for Air	MOE Environmental Quality Standards for Water / Groundwater	Ministry of Environment Japan
India	NAAQS 2009 / CPCB AQI / state PCB	CPCB designated best use; EPA Rules Schedule VI; BIS; state PCB consents	India Code; MoEFCC; CPCB; MPCB / other State PCBs
South Africa / Africa	National standards and provincial/local implementation	National Water Act objectives, water-use licences and national water-quality guidelines	Government gazette; Department of Water and Sanitation; UNEP / FAOLEX
Latin America	Country ministry standards; national monitoring portals	National surface water / potable / effluent rules; basin authorities	National gazettes; FAOLEX; ECOLEX
Middle East / Asia-Pacific	National environment ministries and standards institutes	National water quality, reuse, discharge and marine standards	National gazettes; FAOLEX; ECOLEX

6.1 Global environmental law and standards discovery

- FAOLEX: open repository with national laws, regulations, policies and agreements across 300+ jurisdictions; filter by air, water, environment, waste and other subjects. FAOLEX.
- ECOLEX: information service operated jointly by FAO, IUCN and UNEP; includes treaties, soft-law, national legislation, judicial decisions and literature. ECOLEX.
- National legal gazette / official regulator: final authority for current law, commencement date, amendments, consolidation, exemptions, standards, methods and permit conditions.

For every country added to production, create a “jurisdiction pack”: official environment ministry; national standards institute; air monitoring agency; water monitoring agency; legal gazette; water / wastewater regulator; pollutant release registry; administrative boundary and catchment source. Have a human reviewer validate the pack before enabling compliance verdicts.

7. Global and regional datasets: where to fetch data

Separate observation datasets, modelled products, legal sources, emission inventories and geographic layers.

7.1 Air quality observation and model datasets

Dataset / provider	Coverage and data	Fetch method / use	Important caveat
OpenAQ API v3	Global open ambient air observations; PM2.5, PM10, SO2, NO2, CO, O3 and metadata.	REST API v3: locations, sensors, latest and historic measurements; build country / radius / parameter queries. Docs | API reference.	OpenAQ explicitly says it does not contain every monitoring dataset worldwide. It aggregates available data, so track provider, license, sensor and station metadata. v1/v2 retired; use v3.
WHO Ambient Air Quality Database V8.0 (June 2026)	City/town annual mean PM2.5, PM10, NO2; global comparison and exposure context.	Download workbook or use WHO interactive app. WHO database.	Not a real-time station API; data represent city/town averages, update every 2-3 years and have coverage limitations.
US EPA AQS API / AirData	USA historical monitor-level samples, station metadata, quality data; daily/hourly/annual extracts.	AQS row-level REST API and AirData bulk downloads. AQS API | Obtaining AQS data.	AQS is not real-time and may lag sample collection by six months or more. Use AirNow for real-time public reporting.
AirNow API	USA and participating international partner real-time air-quality observations / forecasts.	Developer API / web services. AirNow API.	Operational/nowcast product; not interchangeable with certified AQS design-value data. Show timestamp and “preliminary” flags.
EEA Air Quality e-Reporting / download service	European air quality monitoring time series, stations and validated/up-to-date reporting from member and cooperating countries.	EEA download service, API, spatial services, zipped Parquet. EEA air-quality download.	Observe reporting status / validation status; monitoring coverage differs by country.
Copernicus CAMS	Global atmosphere composition: analyses, forecasts, reanalysis and recent retrospective records.	Copernicus Data Space / CAMS access; global composition forecasts and analysis. CAMS data docs.	Modelled / assimilated product, not equivalent to a ground station or statutory reference monitor.
Sentinel-5P / TROPOMI	Global satellite atmospheric products, including NO2, SO2, CO, ozone and aerosol indices.	Copernicus Data Space Sentinel-5P products and APIs. Sentinel-5P docs.	Satellite column measurements / retrievals differ from breathing-zone surface concentration. Cloud, resolution and QA filters matter.
Canada NAPS	Canadian ambient air sampling and monitoring time series.	NAPS Data Portal / Data Query Tool. NAPS program.	National Canadian data; use its station/method metadata and applicable CAAQS version.
EDGAR v8.1 Air Pollutant Emissions	Global anthropogenic source emissions by country / sector, including PM2.5, PM10, NOx, SO2, NH3, CO and NMVOC.	Download emissions totals and gridded maps. EDGAR air pollutants.	Emissions inventory, not ambient concentration. Use to discuss likely pressure sources, not local exposure or legal compliance.

7.2 Water-quality observation datasets

Dataset / provider	Coverage and data	Fetch method / use	Important caveat
UNEP GEMS/Water GEMStat	Global freshwater monitoring data, stations, country / catchment summaries and 600+ parameter catalogue.	Portal download; query by station, parameter, date and geography; open-data subset on Zenodo. GEMStat portal | Zenodo open data DOI.	Data shared by countries/organisations; licenses vary. Bulk export may need a request. Portal announced temporary outage on 7 Oct 2026; verify current status.
UNEP SDG 6.3.2 data	Country-level / water-body good ambient water quality indicator and national method metadata.	Use SDG 6.3.2 data, indicator metadata and reporting. UNEP indicator | UNSD metadata.	Not a complete station-level raw chemistry feed; "good" target is set nationally.
USGS Water Data API / Water Quality Portal	USA water quality samples from USGS, EPA and 400+ federal, state, tribal and local agencies.	Use Samples API / Water Quality Portal API, OGC APIs, STAC and downloads. USGS API docs.	Discrete sample data have varying frequency and methods; do not imply continuous monitoring where only grab samples exist.
EEA Waterbase / WISE SoE Water Quality	Europe rivers, lakes, groundwater, transitional, coastal and marine waters; water quality and pollutant pressures.	2026 Waterbase ICM dataset available in CSV/TXT/SQL/SQLite; Discodata query endpoint. Waterbase data page | WISE Freshwater.	2026 Waterbase dataset published 7 Jul 2026 with temporal coverage through 2025. Preserve sampling period and reporting method.
Australia Water Data Online	Thousands of Australian water monitoring stations; levels, flow, EC, turbidity, pH, water temperature, rainfall and storage.	Download time-series data from Bureau of Meteorology Water Data Online. Water Data Online.	Primarily water quantity and selected field-quality parameters; not a full toxicant laboratory database.
WHO / UNICEF JMP WASH database and API	Global country / subnational household, school and health-care-facility drinking-water service-level estimates.	REST API base `https://api.washdata.org/v1`; JSON / CSV, no API key. JMP data | JMP API documentation.	Measures drinking-water service/access, sanitation and hygiene; not raw chemical tests from rivers or tap samples.
FAO AQUASTAT	Global country-level water resources and agricultural water management variables / indicators.	Download / query AQUASTAT. AQUASTAT.	Water-resource and use context, not a substitute for measured pollution concentration.

7.3 Industrial releases, permits and compliance datasets

Dataset	Coverage	Use in feature	Link
US EPA ECHO / ICIS-NPDES	USA regulated facilities, permit limits, discharge monitoring reports, inspections, violations and enforcement.	Identify facility and permit, permitted outfalls, reported effluent, exceedance and enforcement history.	ECHO downloads; ECHO web services
EU Industrial Emissions Portal / E-PRTR	Major industrial sites across Europe; releases / transfers to air, water and land.	Potential source inventory and reported emissions; join by facility ID and location.	Industrial emissions datasets
National PRTR / permit portals	Country-specific point-source reporting and regulatory records.	Search the country legal pack; obtain permits, monitoring reports, inspection records and pollutant releases.	FAOLEX / ECOLEX plus national environment ministry portals

7.4 Satellite and geospatial datasets - useful, but not a lab result

Dataset	What it provides	What it cannot prove alone
JRC Global Surface Water v1.4	Surface-water occurrence, seasonality and transition from Landsat imagery; historic mapping 1984-2021. Dataset details.	Does not measure BOD, COD, metals, pathogens or legal pollution status. Product temporal coverage ends 2021.
Sentinel-2 / Landsat	Optical imagery that can support research estimates of turbidity, chlorophyll-a, suspended matter and algal blooms with calibrated, site-specific models.	Clouds, shallow water, bottom reflectance, coloured dissolved organic matter and calibration limit accuracy. Validate with lab samples.
HydroBASINS / HydroRIVERS / HydroATLAS	Global catchment boundaries, sub-basins and river network for watershed joins and upstream/downstream analysis. HydroBASINS | HydroRIVERS | HydroATLAS.	Hydrological geometry and attributes; not current pollutant monitoring.
geoBoundaries / OCHA CODs	Global country and administrative subdivision boundaries (ADM0-ADM5 depending coverage). geoBoundaries API | OCHA CODs.	For legal district mapping, prefer official national boundaries and keep the source year / license. Global boundaries may be simplified or disputed.

8. District-wise and water-body-wise mapping

The map should show both administrative district and hydrological water-body identity.

8.1 Recommended geospatial joining process

- Ingest the exact monitoring station coordinates, station code, monitor type and source agency.
- Use a point-in-polygon spatial join to attach country, state/province, district/county and local authority from a versioned administrative boundary dataset.
- Use hydrographic layers to attach catchment / sub-basin, river reach, lake polygon, watershed and upstream/downstream context where available.
- If a station lies close to a district boundary, store the join method and confidence; do not silently force a district if coordinate uncertainty overlaps two polygons.
- Preserve original water-body name and official code exactly as supplied by the regulator. Keep aliases / transliterations in separate fields.
- For cross-district rivers, store multiple district associations or station-level district; never propagate one station's result to all districts along the river.

8.2 District-wise dashboard measures

Dashboard measure	Correct calculation	Avoid
Latest monitored reading	Latest valid observation at a station, with timestamp and parameter.	Calling it "district pollution" when one station represents only one point.
District median / range	Aggregate valid station-level results for a defined period, parameter and medium; show n stations and n samples.	Mixing different pollutants, units, methods or sampling dates.
Exceedance frequency	Number / percentage of valid assessment periods above a specified applicable threshold.	Counting individual sensor spikes where the standard uses annual or percentile statistics.
Polluted water-body count	Count distinct officially defined water-body segments with validated assessment status.	Counting multiple samples at the same stretch as separate water bodies.
Trend	Compare same station, parameter, method and season across time; show sample count and confidence.	Comparing different stations or changing monitoring networks without adjustment.
Source pressure	Map upstream facilities, land use, urban areas and discharge points as potential pressures.	Claiming causation from proximity alone.

8.3 Geospatial fields to retain

Store `country_iso3`, `admin1_code`, `admin2_code`, `admin_name_source`, `admin_boundary_version`, `latitude`, `longitude`, `coordinate_accuracy_m`, `station_code`, `waterbody_id`, `waterbody_type`, `river_reach_id`, `subbasin_id`, `basin_id`, `upstream_downstream_position`, `catchment_area_km2` where available, and `spatial_join_method`. For a point-source discharge, separately store facility, permit, outfall and receiving-water-body IDs.

9. What evidence-based answers should explain

A good answer shows the measurement, rule, comparison, source and uncertainty - not just a colour badge.

9.1 Required explanation blocks

Block	What to display
Question / reported issue	User’s location, date, suspected medium and concern. Distinguish a complaint from measured evidence.
Observation evidence	Parameter, value, unit, timestamp, averaging window, sample type, station / coordinates, agency, laboratory / method, QA flags and source URL.
Applicable rule	Jurisdiction, exact instrument, issuing authority, version, legal status, section/table/annex, effective date and whether current on sample date.
Threshold and form	Numeric limit, unit, comparator, averaging period, percentile / frequency form, water class or designated use, test temperature and sample method.
Comparison math	Observed statistic; limit; difference; exceedance ratio; valid-data count; formula. For minimum DO, compare in the opposite direction.
Interpretation	Within / above / below / not assessable; what this can indicate and what it does not prove.
Recommended follow-up	Confirmatory sample, additional parameters, repeat interval, responsible regulator / service, inspection or incident action.
Evidence trail	Clickable source document, exact page/table/paragraph, retrieval timestamp, archived source version and data provenance.

9.2 Output statuses that avoid overclaiming

Status	When to use
Within selected criterion	Valid observation and correct statistical form are available and comparison is within the applicable threshold. This is not a guarantee that all pollutants are safe.
Potential exceedance	An observation exceeds a relevant value, but full legal test conditions, QA or monitoring period may not be satisfied.
Formal non-compliance indicated	Only when all applicable legal requirements, statistical form, sampling protocol and facility / permit context are verified. Human / regulator review may still be required.
Insufficient data	Missing parameter, required averaging period, valid data count, method, water-use class, location or applicable standard.
Guideline comparison only	No local enforceable standard resolved; compare against WHO / technical guideline but label as non-binding.
Unverified report	Citizen report, photograph, field sensor or imported value without confirmatory evidence.

9.3 Example: air-quality explanation

Illustrative scenario only: A monitor reports PM2.5 = 22 µg/m³ as a valid 24-hour mean. The platform should say: “This 24-hour mean is above the WHO 2021 24-hour AQG level of 15 µg/m³. The WHO value is health-based guidance, not necessarily the local legal limit. The local legal assessment is pending because the jurisdiction-specific air standard and completeness rule have not yet been confirmed. This observation alone does not establish an annual-standard violation.”

9.4 Example: river-water explanation

Illustrative scenario only: A river sample has BOD5 = 4.5 mg/L at 20°C. If the water-body stretch is officially designated for an outdoor bathing use with a 3 mg/L BOD5 criterion, the platform can flag a potential exceedance of that use-based criterion. It must not compare the same number with a 3-day-at-27°C effluent standard or infer which upstream facility caused the result.

9.5 Example: facility discharge explanation

Illustrative scenario only: An industrial discharge report shows COD above a numeric permit limit. Retrieve permit ID, outfall, sample date, compliance period, lab method, reported flow, averaging basis and the permit clause. If any of those is missing, say “reported result above permit limit - verification required”, not “facility illegally polluted the river”.

10. Recommended database schema and data pipeline

The core architecture should keep raw observations, legal rules and AI explanations as separate records.

10.1 Core tables

Table	Key fields / purpose
`source_registry`	`source_id`, publisher, source_type, jurisdiction, canonical_url, license, API_type, retrieval_timestamp, refresh_frequency, parser_version, trust_tier, active_flag.
`legal_instruments`	`instrument_id`, title, authority, country/state, instrument_type, official_gazette_ref, legal_status, publication_date, effective_from, effective_to, amendment_version, official_url, archived_file_hash.
`standards`	`standard_id`, instrument_id, medium, sample_type, waterbody_type, designated_use, pollutant_id, numeric_value, comparator, unit, averaging_period, percentile/form, test_method, temperature, salinity/hardness/pH qualifiers, notes, effective dates.
`parameter_dictionary`	Canonical pollutant name, synonyms, CAS / parameter code, chemical form, unit conversion rules, analyte reporting basis, matrix, detection-limit unit.
`observations`	`observation_id`, source_id, station_id, parameter_id, value, unit, sample_date_time, averaging_window, result_status, method, sample_type, QA/QC flags, detection limit, raw record ID.
`stations`	Station code, name, provider, coordinates, location accuracy, operator, monitor type, active dates, station metadata.
`places`	Country, admin1/admin2 codes, district name, boundary dataset version, basin/subbasin, waterbody and river reach IDs.
`facilities_permits`	Facility ID, permit ID, industry, outfall, coordinates, receiving water, permit limit, sampling conditions, start/end date, compliance records.
`assessments`	Observation(s), selected standard, method/form used, status, computed statistic, difference, ratio, data sufficiency, algorithm version, human review.
`answer_evidence`	Assessment ID, source document, exact page/table/paragraph, quotation/paraphrase span, citation URL, content hash, retrieval date, RAG chunk ID.
`incidents_actions`	Complaint, case status, responsible authority, recommended step, assigned owner, action due date, closure evidence and audit history.

10.2 Normalisation rules

- Preserve the original raw value and unit; store a second normalised value only with a transparent conversion record.
- Keep chemical form distinct: nitrate (NO3), nitrate-nitrogen (NO3-N), total nitrogen, ammonium and ammonia are not interchangeable.
- Keep total vs dissolved metals / nutrients; record filtered / unfiltered basis.
- Keep BOD5 at 20°C distinct from BOD3 at 27°C; never convert by a naive unit conversion.
- Keep µg/m³ distinct from mg/m³; keep water units µg/L, mg/L and µg/kg; do not mix concentration with pollutant load.
- For air data preserve local timezone and UTC timestamp. For water samples preserve collection time, depth, tide / flow stage and lab receipt time where available.
- For “less than detection limit” store the censoring indicator and detection limit. Do not treat it as exactly zero.
- For any derived index retain formula version, breakpoint source and rounding rule.

11. Threshold selection and comparison engine

The threshold resolver should be deterministic and explainable; AI is the narrative layer, not the calculator.

11.1 Required inputs to the rule resolver

Input	Examples
Jurisdiction	Country, state/province, district/county, local authority and applicable transboundary basin.
Medium / sample	Ambient air, stack, river, lake, groundwater, coastal water, drinking water, effluent, soil, sediment.
Intended use	Bathing, drinking source, aquatic life, irrigation, industrial cooling, shellfish harvesting, general ambient ecosystem.
Source / activity	Industrial sector, municipal STP, mining, landfill, agriculture, household, transport, natural source.
Date context	Sample date and time; standard effective date; transition / grace period; permit validity.
Measurement semantics	Pollutant and chemical form, unit, method, averaging period, percentile, duration, temperature, flow / salinity / hardness if relevant.

11.2 Rule selection order

- 1. Resolve official place and environmental entity from coordinates / user input.
- 2. Resolve sample type and intended use; ask for missing context or return “insufficient information”.
- 3. Retrieve the current and historic applicable legal instrument(s) from the jurisdiction registry.
- 4. Check facility permit / consent / site-specific direction before general limits.
- 5. Check industry-specific / discharge-specific standards before general fallback standards.
- 6. Check designated water-body class / water use / ecological objective before comparing ambient-water results.
- 7. Validate method, units, statistic and temporal completeness against the standard’s conditions.
- 8. Calculate compliance screening in code; attach the source and exact table / annex citation.
- 9. Use WHO / UNEP / IFC / scientific guidance as a separately labelled contextual comparison.
- 10. If there are conflicting or overlapping laws, return conflict / human review - do not have the LLM select the convenient threshold.

11.3 Comparison mathematics

Upper limit: exceedance ratio = measured statistic / permitted maximum. Difference = measured statistic - limit. Flag if ratio > 1, subject to the standard’s method and permitted exceedance form.
Lower limit (e.g. dissolved oxygen): shortfall = minimum required - measured statistic. Flag if measured statistic < minimum required.
Range requirement (e.g. pH): compare against both lower and upper bound; outside if value < minimum or value > maximum.
Percentile / frequency requirement: calculate the prescribed distribution / count over the official assessment window; do not substitute the latest sample.

11.4 Pseudocode

```
assessment = resolve_context(observation, coordinates, waterbody, intended_use) if not assessment.context_complete: return INSUFFICIENT_DATA(assessment.missing_fields) rules = standards_registry.find( jurisdiction=assessment.jurisdiction, medium=assessment.medium, sample_type=assessment.sample_type, intended_use=assessment.intended_use, valid_on=observation.sample_date ) rule = rank_by_legal_applicability(rules, permit_first=True, sector_specific_before_general=True) if rule is None: return GUIDELINE_ONLY(compare_with_WHO_or_reference_guidance()) validity = validate_method_units_period_and_completeness(observation, rule) if not validity.ok: return INSUFFICIENT_DATA(validity.reasons) statistic = calculate_required_form(observation.series, rule.form) result = compare(statistic, rule.operator, rule.value, rule.bounds) return create_assessment(result, rule_id=rule.id, formula_version=rule.formula_version, source_citations=rule.exact_citations)
```

12. RAG and evidence-citation design

Use retrieval to explain the applicable rule; do not ask the language model to invent or calculate thresholds.

12.1 Recommended evidence pipeline

- Document ingestion: download official Acts, rules, standards, permits, monitoring protocols, annual reports, annex tables and data dictionaries.
- Metadata extraction: issuer, jurisdiction, legal status, publication/effective dates, amendment number, medium, pollutant, table/section and source version.
- Table-aware extraction: preserve row and column headers, units, footnotes, qualifiers, methods and exceptions. A threshold without its unit / averaging period is not a usable rule.
- Versioning: save original PDF/HTML, retrieval time, content hash, superseded date and parser version. Never overwrite an old law when a new amendment appears.
- Structured rule load: a reviewed human-approved threshold record is the source for calculations. RAG chunks are the evidence narrative around the record.
- Answer generation: pass the deterministic result, source snippets, citation metadata and uncertainty flags to the LLM. Require claims to be supported by retrieved evidence.
- Audit: save input data, selected rule ID, computed result, prompt/model version, source IDs and user-visible answer.

12.2 Suggested retrieval filters

Filter first by country / state and legal validity date; then by environmental medium; then by sample type / designated use / sector; then by pollutant / parameter; then search the specific rule, schedule, annex, table, method and exceptions. Semantic similarity alone is unsafe for selecting legal limits.

12.3 AI answer contract / JSON shape

```
{
 "assessment_status": "potential_exceedance | within_criterion | insufficient_data | guideline_only",
 "jurisdiction": {"country": "...", "subdivision": "..."},
 "observation": {"parameter": "...", "value": 0, "unit": "...", "sample_date": "...", "sample_type": "..."},
 "applicable_rule": {"instrument": "...", "issuer": "...", "legal_status": "binding | guidance",
                     "section_table": "...", "effective_from": "...", "source_url": "..."},
 "threshold": {"operator": "<=", "value": 0, "unit": "...", "averaging_period": "...", "method": "..."},
 "comparison": {"observed_statistic": 0, "difference": 0, "ratio": 0, "validity_checks": []},
 "explanation": "Plain-language, scoped explanation",
 "limitations": [],
 "next_steps": [],
 "citations": [{"url": "...", "page": 0, "table": "...", "retrieved_at": "..."}]
}
```

Do not let the LLM choose its own numeric threshold from loose semantic search results. The application should resolve and validate the structured rule first, then pass the selected record and supporting legal excerpt to the LLM.

13. Data quality, uncertainty, risk and governance

Environmental decisions depend on measurement quality and legal context as much as on the numeric value.

13.1 Minimum data-quality checks

Quality dimension	Check
Identity	Station / facility ID, coordinates, sample ID, water-body or monitor identity and provider.
Time	Sample collection time, timezone, analysis time, record update time and applicable legal date.
Method	Recognised method, accreditation / certification where applicable, instrument ID, calibration and lab QA/QC.
Completeness	Required hourly / daily / annual coverage, replicate count, sampling frequency and missing period.
Units / matrix	Correct physical unit, chemical species, dissolved/total basis, sample matrix and detection limit.
Plausibility	Outlier check, impossible values, duplicate records, sensor drift, station relocation and method change.
Provenance	Raw source, license, data owner, retrieval timestamp, transformations and hash.
Representativeness	Station type, distance from user, upstream/downstream position, water depth, tide / flow, weather and seasonal context.

13.2 Measurement uncertainty

Show the monitoring method and uncertainty where the source publishes them. Low-cost sensors, remote sensing and citizen observations can be useful for screening, but their uncertainty, calibration and QA flags should affect confidence. If a result lies close to a threshold, do not force an overconfident pass/fail label. Display the difference, uncertainty or data-quality caveat and recommend confirmatory monitoring when appropriate.

13.3 Causal attribution

A pollution exceedance at a river station does not identify who caused it. To attribute a source, require additional evidence such as upstream/downstream sampling, discharge records, flow and pollutant load, chemical fingerprints, timing, source inspection and regulator findings. Facility proximity, an industrial cluster or a satellite image is not enough to assign liability.

13.4 Public-facing language

- Use “measured above the selected criterion” rather than “illegal pollution” unless a legal non-compliance determination is supported.
- Use “potential source / pressure” rather than “polluter” when source attribution is unverified.
- Show data age: “sampled on [date]”; never present a historical reading as live conditions.
- Show the geographic scope: “station X on stretch Y”, not “whole district”, unless a statistically valid district assessment exists.
- Show missing data: “not monitored / no recent result available” is not the same as “clean”.

14. Implementation roadmap and acceptance checklist

Build a smaller verified jurisdiction pack first, then expand to new countries.

14.1 Suggested phased implementation

Phase	Deliverable	Exit condition
1 - Global schema	Canonical pollutants, units, sample types, jurisdiction/place model, source registry and legal-rule schema.	Can distinguish ambient air, ambient water, drinking water and effluent.
2 - Pilot geography	One country and one subnational regulator (recommended: India/Maharashtra first), plus one global reference layer (WHO).	Every demo answer shows exact source and data provenance.
3 - Air ingestion	CPCB / national data, OpenAQ fallback, WHO city data, optional satellite/model layers.	Local AQI formula separated from WHO comparison; historical and real-time labels correct.
4 - Water ingestion	CPCB/MPCB / national water monitoring plus GEMStat/EEA/US WQP coverage where permitted.	Water-body IDs and sample types preserved; no blanket river/district inference.
5 - Standards engine	Versioned rule registry, jurisdiction resolver, effective dates, method/averaging validation.	Test fixtures cover limits, minimum values, ranges, percentiles and insufficient data.
6 - RAG explanation	Citation-aware answer generator with page/table references and refusal to infer unsupported facts.	Every numeric limit resolves to a reviewed structured rule and primary legal source.
7 - Incident workflow	Case status, authority routing, corrective action, repeat sampling and audit trail.	A reported issue can be tracked from initial complaint through evidence and closure.
8 - Expand global coverage	Add jurisdiction packs for EU, USA, Canada, Australia/NZ, China, Japan and additional countries.	Each pack verified by human review with dated source records.

14.2 Acceptance checklist

- [ ] Can the system tell an air-quality concentration from an AQI value?
- [ ] Can it state which country/region AQI formula it used?
- [ ] Can it distinguish WHO guidance from binding local law?
- [ ] Can it distinguish ambient river water from drinking water and effluent?
- [ ] Can it select a water-body designated use or explicitly say it is unknown?
- [ ] Can it match averaging period, percentile, method and chemical form?
- [ ] Can it detect old, incomplete, invalid or unverified observations?
- [ ] Can it identify the exact law version effective on the sample date?
- [ ] Can it show the original source URL, section/table/page and retrieval date?
- [ ] Can it say “insufficient data” rather than guessing?
- [ ] Can it avoid claiming a responsible polluter without source-attribution evidence?
- [ ] Can it show a station or water-body stretch rather than overgeneralising to a district?
- [ ] Can a reviewer reproduce the numeric comparison from stored data and rule ID?

15. Official reference directory

Use primary sources for laws and standards; use aggregator datasets for discovery and coverage.

15.1 Global standards and legal discovery

WHO Global Air Quality Guidelines (2021) - https://www.who.int/publications/i/item/9789240034228
Health-based levels for six key air pollutants; not automatically binding law.

WHO Drinking-water Guidelines (2026) - https://www.who.int/publications/i/item/9789240121225
Latest located edition: fourth edition with first, second and third addenda.

UNEP SDG 6.3.2 - https://www.unep.org/en/goal-6-clean-water-and-sanitation/indicator-632
Global ambient freshwater quality indicator and core parameters; targets set nationally.

UNEP GEMS/Water GEMStat - https://gemstat.org/data-gemstat/data-portal/
Global freshwater station/country/catchment datasets; licensing and access vary.

FAOLEX - https://faolex.fao.org/
National environmental and resource laws / policy repository across jurisdictions.

ECOLEX - https://www.ecolex.org/p/about/
Environmental treaties, legislation, soft law, cases and literature.

IFC / World Bank Group EHS - https://www.ifc.org/en/insights-reports/general-environmental-health-and-safety-guidelines
Industrial environmental health, air, water, wastewater, waste and monitoring guidance.

UN Sustainable Development Goal metadata - https://unstats.un.org/sdgs/metadata/?Goal=6&Target;=6.3&Text;=
Official indicator definitions, methods and reporting metadata.

15.2 Air data and law sources

OpenAQ API v3 docs - https://docs.openaq.org/about/about
Global aggregated ground-level observations; not exhaustive; v3 only.

WHO Air Quality Database - https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database
City-level annual mean PM2.5, PM10 and NO2; 2026 V8 update.

US EPA AQS API - https://aqs.epa.gov/aqsweb/documents/data_api.html
Historical row-level US ambient-air samples; not real-time.

AirNow API - https://docs.airnowapi.org/
Real-time AQI and forecast product for supported geographies.

EEA Air Quality Download Service - https://www.eea.europa.eu/en/datahub/datahubitem-view/778ef9f5-6293-4846-badd-56a29c70880d
EU/EEA air-monitoring time-series and station information.

Copernicus CAMS - https://documentation.dataspace.copernicus.eu/Data/CopernicusServices/CAMS.html
Global atmosphere composition analysis, forecast and reanalysis.

Sentinel-5P - https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel5P.html
Satellite atmospheric trace-gas / aerosol products.

EDGAR AP v8.1 - https://edgar.jrc.ec.europa.eu/dataset_ap81
Global air-pollutant emissions inventories by sector/country; not ambient concentrations.

US EPA NAAQS - https://www.epa.gov/criteria-air-pollutants/naaqs-table
Current US ambient-air standards and their forms.

EU Ambient Air Directive 2024/2881 - https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024L2881
EU ambient-air limit/target values and transition dates.

China GB 3095-2026 announcement - https://www.mee.gov.cn/xxgk2018/xxgk/xxgk01/202602/t20260224_1144401.shtml
Chinese ambient-air standard effective 1 March 2026.

15.3 Water data and law sources

GEMStat Zenodo repository - https://doi.org/10.5281/zenodo.13881899
Open-licensed subset from GEMS/Water, annual update; preserve license attribution.

USGS Water Data API - https://api.waterdata.usgs.gov/docs/
USGS samples and Water Quality Portal access.

US Water Quality Portal - https://www.waterqualitydata.us/
Discrete samples from USGS, EPA and partner agencies.

EEA Waterbase Water Quality ICM 2026 -
https://www.eea.europa.eu/en/datahub/datahubitem-view/fbf3717c-cd7b-4785-933a-d0cf510542e1
European water-quality datasets, 1900-2025 coverage; CSV/SQLite/Discodata.

WISE Freshwater - https://water.europa.eu/freshwater
Water Framework Directive data, pressures, water-body status and measures.

US EPA Water Quality Criteria - https://www.epa.gov/wqc/national-recommended-water-quality-criteria-tables
Recommended aquatic-life and human-health criteria for many pollutants.

CCME Canadian Environmental Quality Guidelines - https://ccme.ca/en/current-activities/canadian-environmental-quality-guidelines
Chemical-specific guideline values and supporting documents.

Australia/New Zealand Water Quality Guidelines - https://www.waterquality.gov.au/guidelines/anz-fresh-marine
Fresh/marine ecosystem framework and toxicant default guideline values.

Japan Environmental Quality Standards - https://www.env.go.jp/en/standards/
Japan water, groundwater and air standards.

China GB 3838 surface water - https://english.mee.gov.cn/Resources/standards/water_environment/quality_standard/200710/t20071024_111792.shtml
Surface water quality classes and national standard reference.

EU Drinking Water Directive 2020/2184 - https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:32020L2184
EU parametric values for water intended for human consumption.

EU WFD and 2026 amendment - https://environment.ec.europa.eu/topics/water/water-framework-directive_en
EU water-body status framework and river basin planning.

EU Directive 2026/805 - https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32026L0805
2026 amendment to WFD, groundwater and environmental quality standards.

EU Urban Wastewater Directive 2024/3019 - https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:32024L3019
Recast municipal wastewater collection and treatment duties.

WHO/UNICEF JMP data API - https://washdata.org/data/api
Global drinking-water access / WASH estimates, not raw pollutant samples.

FAO AQUASTAT - https://www.fao.org/land-water/resources/tools/databases/aquastat/en
Country-level water resources and agricultural water management.

15.4 Facilities and spatial reference sources

US EPA ECHO downloads - https://echo.epa.gov/tools/data-downloads
US regulated facilities, permits, discharge monitoring and violations.

EU Industrial Emissions dataset - https://industry.eea.europa.eu/industrial-emissions/dataset
Industrial sites and reported releases/transfers to air, water and land.

HydroBASINS - https://www.hydrosheds.org/products/hydrobasins
Global sub-basin boundaries.

HydroRIVERS - https://www.hydrosheds.org/products/hydrorivers
Global river network and reach IDs.

HydroATLAS - https://www.hydrosheds.org/hydroatlas
Global basin, river and lake hydro-environmental attributes.

geoBoundaries API - https://www.geoboundaries.org/api.html
Global administrative boundaries; inspect year, source and license.

UN OCHA COD datasets - https://knowledge.base.unocha.org/wiki/spaces/imtoolbox/pages/42045911/Common%2BOperational%2BDatasets%2BCODs
Country administrative boundaries and P-codes, especially humanitarian contexts.

JRC Global Surface Water - https://developers.google.com/earth-engine/datasets/catalog/JRC_GSW1_4_GlobalSurfaceWater
Historic surface-water presence/change mapping; not pollution chemistry.

15.5 India-specific starting references retained from the project baseline

CPCB - https://cpcb.nic.in/
AQI, NAAQS, water-quality criteria, polluted river stretch assessments.

MPCB Water Quality - https://www.mpcb.gov.in/en/water-quality
Maharashtra monitoring network, bulletins and coastal water-quality documents.

India Code - https://www.indiacode.nic.in/
Primary central legislation and amendments.

MoEFCC - https://moef.gov.in/
Environmental acts, rules, notifications and wetland guidance.

BIS - https://www.bis.gov.in/
Indian drinking-water specification and national standards.

Closing implementation note

For the first production release, do not claim worldwide legal compliance coverage. Launch a verified coverage matrix: countries and subnational areas for which you have (1) current primary legal texts, (2) reviewed structured threshold tables, (3) reliable monitoring data, (4) station and boundary metadata, and (5) a test suite. For every other location, offer an evidence-backed guideline comparison or clearly state that local legal coverage is not yet available.